

FOR CSE, AIE AND CYBER SECURITY TEAMS — A SOFTWARE PRODUCT

A worked sample. Replace every word with your own project. Notice how much of the answer is carried by a figure rather than by a sentence.

A. PROJECT IDENTIFICATION

Project ID	PE-2026-CYS-027
Project Title	PhishGuard: Real-Time Phishing Detection with Explainable Verdicts
Domain	Blockchain and Cybersecurity
Project Category	Software
Department & Programme	Computer Science and Engineering (lead), with Cyber Security and AIE
Project Origin	Independent work, extended from a Machine Learning course project

B. TEAM DETAILS

#	Name of the Student	Roll Number	Programme / Year / Department	Mobile & E-mail
1	A. Nithyasri (Team Leader)	CH.EN.U4CSE22019	B.Tech CSE, III Year	9xxxxxxxxx
2	M. Rohit Kumar	CH.EN.U4CYS22007	B.Tech Cyber Security, III Year	9xxxxxxxxx
3	P. Divya Bharathi	CH.EN.U4AIE23022	B.Tech AIE, II Year	9xxxxxxxxx

C. FACULTY MENTOR

Name & Designation	Dr. K. Vasanthi, Assistant Professor (Sl. Gr.)
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D. ABSTRACT AND KEYWORDS

Phishing pages are the entry point for most credential theft, and the blocklists that browsers rely on only catch a page once somebody has already reported it. This project is a working service that judges a link at the moment it is opened, without waiting for a report. A browser extension sends a hash of the URL to a hosted service, which pulls 46 features from the address itself, from the domain registration record and from the certificate, and passes them to a gradient-boosted classifier. Verdicts are cached for sixty seconds so repeated links return immediately. What separates this service from a plain blocklist is that every verdict is returned with the five features that drove it, so the user is told the domain is nine days old and the certificate does not match, rather than simply being blocked. The service was evaluated on a held-out set of 14 500 URLs, split by time so that no training URL post-dates a test URL. It reached 97.1 per cent accuracy at a 1.8 per cent false positive rate, with a median response of 92 milliseconds.

Keywords (4–6): phishing detection; explainable machine learning; browser security; gradient boosting; zero-day URLs

E. PROBLEM AND OBJECTIVES

Browser blocklists are reactive: a phishing page is blocked only after someone reports it, which typically takes several hours, and most phishing pages are taken down within a day. Students on this campus received 34

credential-harvesting mails in the last academic year that were not blocked by any browser at the time they arrived. Existing commercial tools also return a bare verdict with no reason, so a user who disagrees has nothing to check.

Objectives

- Judge a URL in under 150 milliseconds, end to end, from the browser.
- Reach at least 95 per cent detection accuracy on a time-split held-out set.
- Hold the false positive rate below 3 per cent, since blocking a genuine page is costly.
- Return the five features responsible for every verdict, not just a score.
- Detect at least 85 per cent of URLs first seen after the training cut-off date.

F. DESIGN AND TECHNICAL APPROACH

The service is built in four tiers. A Chrome extension intercepts navigation and sends a hash of the URL to a FastAPI gateway, which handles authentication and rate limiting. A feature extractor pulls 46 features in three groups: lexical features from the URL string, registration age and registrar from the WHOIS record, and issuer and validity from the TLS certificate. These go to a gradient-boosted classifier, chosen over a deep model because it trains on a laptop, runs in nine milliseconds and — more importantly — exposes feature attributions that can be shown to a user. A character-level CNN runs as a fallback where WHOIS lookup fails. Verdicts are cached in Redis for sixty seconds, which matters because a page load fires several requests to the same domain. The explainability service returns the top five contributing features using SHAP values.

SYSTEM ARCHITECTURE

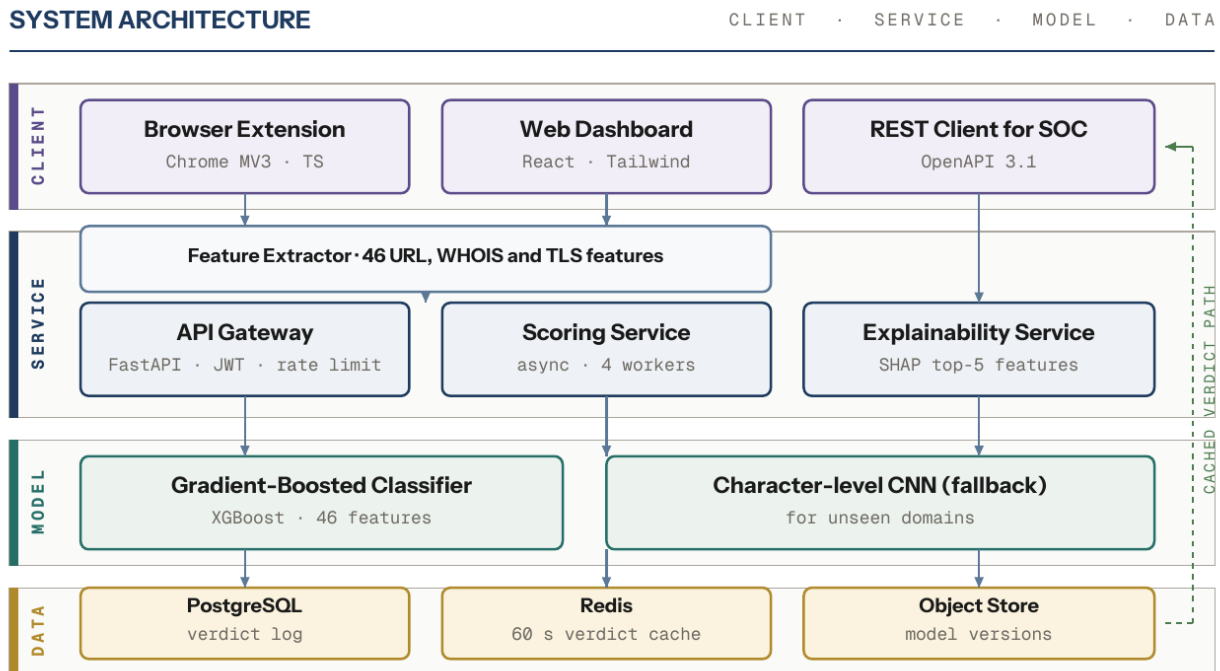


Figure 1 – Four-tier architecture. Cached verdicts bypass the model and return in under 40 ms.

HOW ONE URL IS SCORED

ONE REQUEST

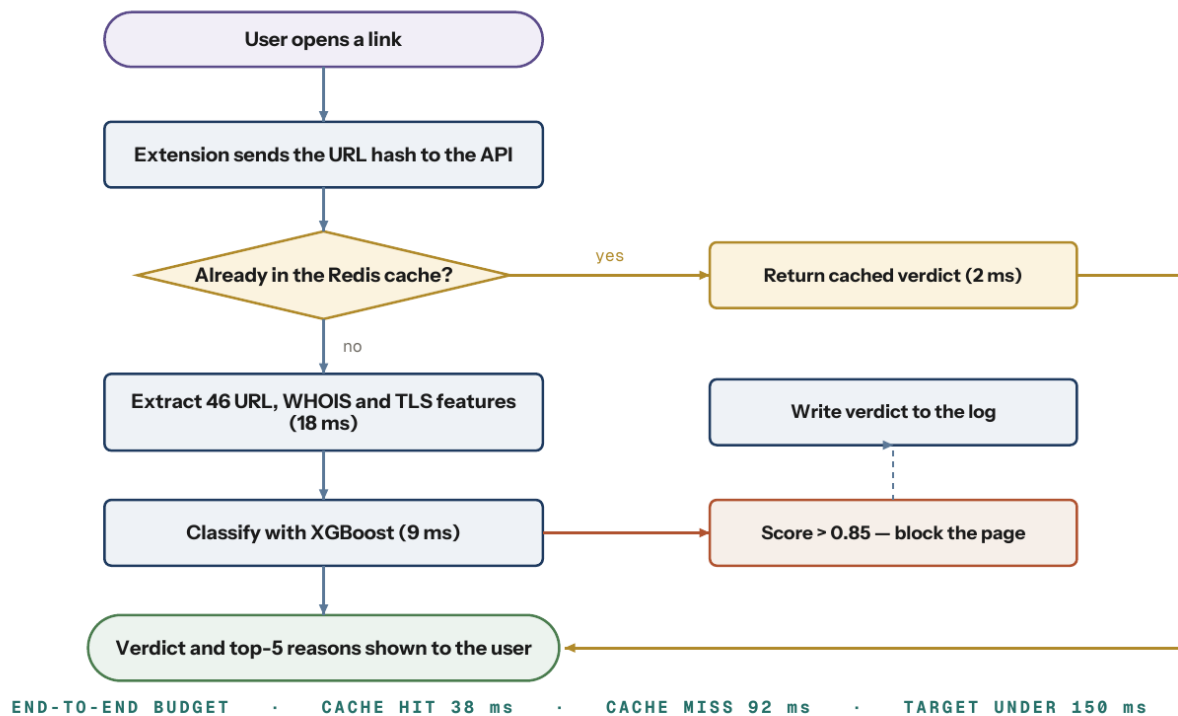


Figure 2 – Request path. Times in brackets are measured medians over 10 000 requests.

MODEL PIPELINE

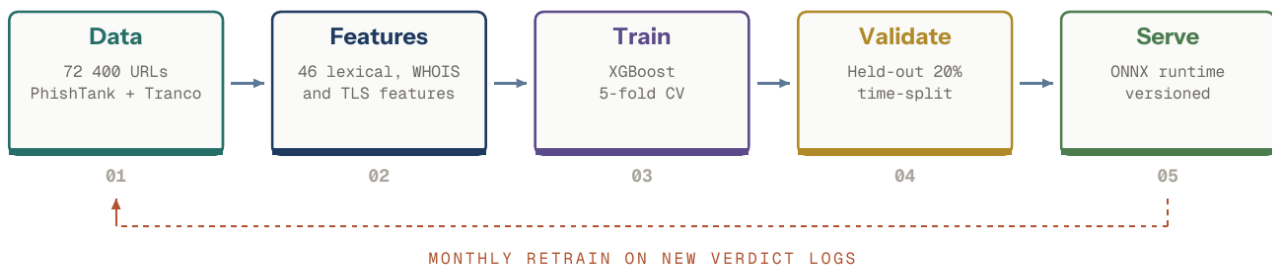


Figure 3 – Training and serving pipeline. Time-split validation prevents look-ahead leakage.

Tools and technologies used: Python, FastAPI, XGBoost, scikit-learn, SHAP, ONNX Runtime, PostgreSQL, Redis, React, TypeScript, Docker, GitHub Actions.

G. IMPLEMENTATION AND RESULTS

The feature extractor and the classifier are complete, tested and exported to ONNX for serving. The API gateway with authentication, rate limiting and the Redis cache is deployed on a campus virtual machine and has served 10 000 test requests. The Chrome extension is in beta and works on Chrome and Edge; the Firefox port is not started. The explainability service returns the top five features but the plain-English wording of those features is still being written, so at present it returns feature names rather than sentences. The SOC dashboard shows the verdict log but the weekly report export is incomplete. Unit-test coverage across the shipped modules stands at 89 per cent. The service can be demonstrated live end to end today.

MODULE BUILD STATUS

OVERALL 89%

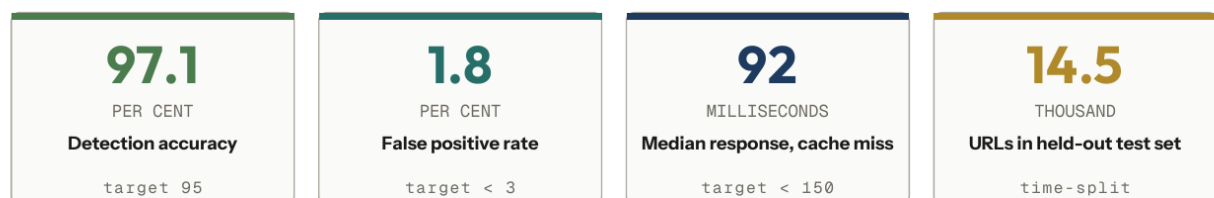
Feature extractor (46 features)		100%	TESTED
XGBoost classifier and ONNX export		100%	TESTED
FastAPI gateway, auth and rate limit		100%	TESTED
Chrome extension (MV3)		95%	BETA
Explainability service (SHAP)		80%	IN PROGRESS
SOC dashboard and reporting		60%	IN PROGRESS

Figure 4 – Module status. Percentages are unit-test coverage against the planned feature set.

Results. *Measured on the dates shown in Figure 5.*

What you measured	Target	Achieved
Detection accuracy (14 500 URL held-out set)	≥ 95%	97.1%
False positive rate	≤ 3%	1.8%
Recall on URLs after training cut-off	≥ 85%	91.4%
Median response, cache miss	≤ 150 ms	92 ms
Median response, cache hit	—	38 ms

MEASURED PERFORMANCE



TARGET VERSUS ACHIEVED



Figure 5 – Evaluated on a 14 500-URL held-out set, time-split so that no training URL post-dates a test URL.

H. WHO IT HELPS

The immediate user is the campus IT security team, who currently learn of a phishing campaign only when a student reports it. The service is packaged as a container and an OpenAPI endpoint, so any institution can host it without sharing browsing data with an outside vendor — which is what makes it usable in a college at all. The explainable verdict also turns each block into a teaching moment for the student. A departmental pilot with the IT team is planned for the coming semester.

I. REFERENCES

- [1] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in Proc. ACM SIGKDD, 2016, pp. 785–794.
[2] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in Proc. NeurIPS, 2017.
[3] A. K. Jain and B. B. Gupta, "A Survey of Phishing Attack Techniques, Defence Mechanisms and Open Research Challenges," Enterprise Information Systems, vol. 16, no. 4, 2022.
[4] OWASP Foundation, OWASP Top 10 Web Application Security Risks, 2021.

Signature of the Team Leader Date:	Signature of the Faculty Mentor Date:
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J. FOR OFFICE USE ONLY

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Evaluation Criterion	Max	Awarded	Remarks of the Jury
Problem and Objectives (Section E)	5		
Design and Technical Approach (Section F)	10		
Implementation and Results (Section G)	15		
Live Demonstration and Performance (Section H)	10		
Application Value (Section I)	5		
Presentation and Documentation (whole sheet)	5		
Total	50		

Screening decision	☐ Shortlisted ☐ Waitlisted ☐ Not shortlisted
Recommended for	☐ Award category: _____ ☐ Compendium ☐ IP support ☐ Incubation
Jury signature & date	